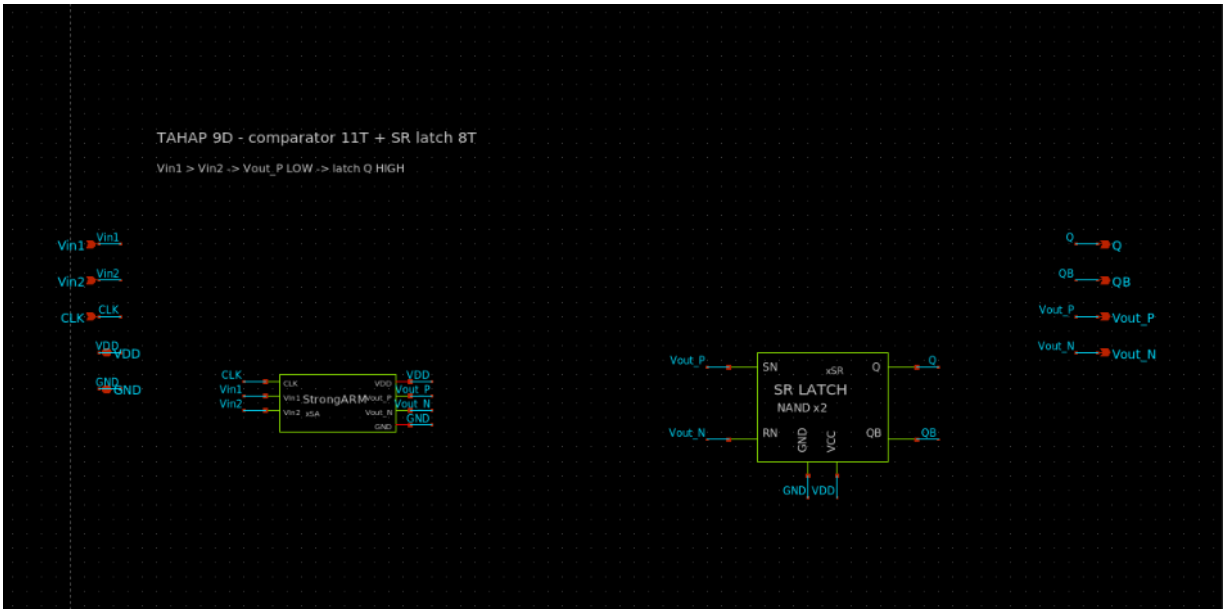
DESIGN NOTE / BLOCK 03

From Regenerative Decisions to Retained Digital State

Design, Characterization, and Integration of an 11-Transistor StrongARM Comparator with an 8-Transistor NAND SR Latch

SKY130 1.8 V / Xschem + ngspice / 4-MHz clock



Evidence-based conclusion. Schematic-level functional verification passed: StrongARM 90/90 PVT cases, NAND2 45/45, SR latch 45/45, integrated circuit 45/45, and final audit 5/5 critical cases. High-voltage device reliability, post-integration mismatch, parasitics, and post-layout performance are not signed off.

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Abstract

This report documents the implementation of a high-speed decision path, from an 11-transistor dynamic StrongARM comparator to a persistent digital output held by an 8-transistor, cross-coupled NAND SR latch. The circuit uses the SKY130 1.8-V MOS models and was characterized through Xschem/ngspice. Analytical relationships, numerical design calculations, simulation methodology, measurements, and limitations are deliberately distinguished.

At a 4-MHz clock with 5-ns edges, the standalone comparator passed 90/90 PVT functional cases; the NAND2 and SR latch each passed 45/45 cases; the integrated design passed 45/45; and five critical configurations passed the final audit. Worst-case integrated decision delay was 3.18913 ns and maximum mean energy was 1.230635 pJ per steady-state cycle. Theoretical predictions agree with observed Boolean behavior, retained state during reset, the approximate logarithmic trend of regeneration delay, and the energy–power identity. In contrast, absolute delay cannot be independently predicted without extracted g_m and internal capacitances; 1.90-V operation retains a documented device-model excursion. The supported claim is therefore *pre-layout schematic-level functional verification*, not silicon-reliability signoff.

1 From a one-bit decision to a usable quantizer block

Block 3 bridges the analog and digital domains of a second-order switched-capacitor delta-sigma modulator. A dynamic comparator identifies the sign of a differential input, but its internal output nodes return to the precharge state during clock reset. Consequently, the comparator alone does not provide a continuously valid stored bit to the downstream circuit. A cross-coupled NAND SR latch remedies this: the decision is captured during evaluation and the preceding state is preserved during reset.

The original system targets are $f_s = 4\,\text{MHz}$, $OSR = 80$, and nominal supply $V_{\text{DD}} = 1.8\,\text{V}$. For a low-pass modulator under the Nyquist convention,

$$T_s = \frac{1}{f_s} = \frac{1}{4 \times 10^6} = 250\,\text{ns}, \qquad BW = \frac{f_s}{2\,OSR} = \frac{4 \times 10^6}{2(80)} = 25\,\text{kHz}. \tag{1}$$

The 250-ns period underlies the conversion between average power and energy per decision. The applied clock has 5-ns rising and falling edges and a source pulse width of 125 ns. These are stimulus specifications: usable output time is established from the actual transient simulation.

The workflow includes standalone StrongARM verification (Stages 1–8), NAND2 qualification (9A), standalone SR-latch verification (9B–9C), integration (9D), and final continuous-HOLD and cycle-energy audit (9E). Each qualification step uses exported Xschem netlists rather than ideal Boolean components alone.

2 Transistor topology and device selection

2.1 Eleven-transistor StrongARM: differential path, regeneration, and reset

The comparator comprises five NMOS and six PMOS devices. Input transistors M1/M2 form the differential transconductor; M7 implements the clock-controlled tail path; M3/M4 and M5/M6 form the regenerative cross-coupled networks. Four PMOS devices S1–S4 provide initialization/precharge of internal and output nodes. All transistors use the SKY130 1.8-V device family; PMOS bodies connect to supply and NMOS bodies to ground.

2.2 Why sizing is a tested engineering choice, not an exact closed-form solution

Geometry W/L governs drive strength and device capacitance, but no single equation uniquely determines eleven dimensions without the noise budget, mismatch targets, internal capacitances, and the actual operating point. These sizes are *selected design values verified numerically*, not a proven global mathematical optimum. The test netlist uses $n_f = 1$ and `mult=1`.

The input pair has $W/L = 10/0.5 = 20$, larger than the cross-coupled NMOS pair ($2/0.3 \approx 6.67$), as an initial choice for a responsive input transconductor. A wider input pair also increases gate loading and can worsen kickback, so increased width is not unconditionally beneficial.

2.3 Eight-transistor latch assembled from two NAND2 gates

Each CMOS NAND2 has two parallel PMOS and two series NMOS devices, each with $W = 2\,\mu\text{m}$ and $L = 0.15\,\mu\text{m}$. Two identical NAND gates form an eight-device SR latch. The integrated hierarchy therefore contains $11 + 2(4) = 19$ MOSFET instances. An individual NAND device has a geometry ratio $W/L = 13.33$. These were initial balanced dimensions subsequently validated functionally over 45 PVT cases, without additional resizing.

Devices	Type	N	W (μm)	L (μm)	W/L	Role
S1–S4	PMOS	4	1.50	0.15	10.00	precharge
M5, M6	PMOS	2	5.00	0.30	16.67	regenerative
M1, M2	NMOS	2	10.00	0.50	20.00	input pair
M3, M4	NMOS	2	2.00	0.30	6.67	regenerative
M7	NMOS	1	1.00	0.30	3.33	tail switch

Table 1: Tested 11T sizing.

Block	PMOS	NMOS	Total
StrongARM	6	5	11
NAND2	2	2	4
SR Latch (2 NAND2)	4	4	8
Integrated	10	9	19

Table 2: MOSFET counts within the tested hierarchy.

3 StrongARM theory: from differential current to regeneration

3.1 Reset phase and evaluation phase

With CLK low, the tail NMOS is off and the PMOS precharge paths bring the comparator outputs HIGH. The active-low SR-latch inputs are therefore both HIGH, indicating HOLD. During evaluation, the input pair creates a current imbalance that can be approximated as

$$\Delta I_D \equiv I_{D1} - I_{D2} \simeq g_{m,\text{in}}\,\Delta V_{\text{in}}, \qquad \Delta V_{\text{in}} = V_{in1} - V_{in2}. \tag{2}$$

This is a local small-signal approximation around a common operating point, not an exact large-signal MOS expression throughout the clocked transient.

For a differential buildup interval t_b, charge conservation provides the approximate initial imbalance

$$\Delta V_0 \simeq \frac{1}{C_{\text{eff}}} \int_0^{t_b} \Delta I_D(t)\,dt \approx \frac{g_{m,\text{in}}\,\Delta V_{\text{in}}\,t_b}{C_{\text{eff}}}. \tag{3}$$

Here C_{eff} includes participating internal node capacitances and *must not* be equated to the externally applied 20-fF output-load capacitor.

3.2 Exponential growth and decision delay

A linearized differential model around the symmetric metastable state yields [1]

$$C_{\text{eq}}\frac{d(\Delta V)}{dt} = g_{m,\text{reg}}\,\Delta V, \qquad \Delta V(t) = \Delta V_0 e^{t/\tau}, \qquad \tau = \frac{C_{\text{eq}}}{g_{m,\text{reg}}}. \tag{4}$$

For an effective logic decision threshold V_L, a first-order delay model is

$$t_{\text{dec}} \approx t_b + \tau \ln\!\left(\frac{V_L}{|\Delta V_0|}\right) \quad \Rightarrow \quad t_{\text{dec}} \text{ tends to increase as } |\Delta V_{\text{in}}| \text{ decreases.} \tag{5}$$

Actual g_m and capacitances vary during evaluation. Thus τ here is an effective approximation, not a precisely known parameter deducible from the W/L table alone.

3.3 Input-referred offset and mismatch

The effective differential input is $(\Delta V_{\text{in}})_{\text{eff}} = \Delta V_{\text{in}} - V_{\text{OS}}$. A Pelgrom-like first-order mismatch scaling is [4][1]

$$\sigma_{\Delta V_T} \propto \frac{A_{V_T}}{\sqrt{WL}}, \qquad \sigma^2_{V_{\text{OS}}} \approx \sigma^2_{\Delta V_T,\text{eq}} + \sigma^2_{\text{other}}. \tag{6}$$

Therefore, a deterministic decision at ± 1 mV does not establish yield across mismatch realizations. An earlier, reported 200-seed local Monte Carlo campaign gave approximately $+0.443$ mV mean offset and 2.471 mV standard deviation. Its per-seed CSV is *not available within the present report archive*; the values are explicitly retained as previously reported, not independently re-audited, results.

4 NAND2 and SR latch: deriving the state-retention function

4.1 First-order pull-up/pull-down balance

In the simplest switch-resistance model, $R_{\text{on}} \propto L/W$. Two identical series NMOS devices give

$$R_{\text{PD,eq}} \approx R_n + R_n = 2R_n, \qquad \left(\frac{W}{L}\right)_{\text{PD,eff}} \approx \frac{1}{2}\left(\frac{W}{L}\right)_n. \tag{7}$$

The parallel PMOS network allows one branch to pull the output HIGH. Lower hole mobility and bias-dependent resistance mean equal 2/0.15-μm PMOS/NMOS widths do not imply symmetric rising and falling delay; that asymmetry is measured rather than assumed.

4.2 Boolean equations and the HOLD state

Let S_N and R_N be active-low inputs. The cross-coupled equations are

$$Q = \overline{S_N \cdot \overline{Q}}, \tag{8}$$

$$\overline{Q} = \overline{R_N \cdot Q}. \tag{9}$$

For $(S_N, R_N) = (0, 1)$, $Q = 1$ (SET). For $(1, 0)$, $Q = 0$ (RESET). For $(1, 1)$, the feedback loop retains the existing state. For $(0, 0)$, the outputs are both forced HIGH and are no longer complements, making this state forbidden during normal operation.

[1] Specific Pelgrom coefficients for the simulated device set were not extracted from the data archive.

The integrated pin mapping is

$$S_N \leftarrow V_{out,P}, \qquad R_N \leftarrow V_{out,N}, \tag{10}$$

with $V_{in1} > V_{in2} \Rightarrow V_{out,P} \approx 0 \Rightarrow Q \approx V_{\mathrm{DD}}$. Once both comparator outputs return HIGH in reset, the latch theoretically holds its last decision. Stages 9D and 9E test this physical transistor-level implication explicitly.

S_N	R_N	Q	$\overline{Q}$	State
0	1	1	0	SET
1	0	0	1	RESET
1	1	Previous Q	Previous QB	HOLD
0	0	1	1	FORBIDDEN

Table 3: Active-low NAND SR latch truth table.

5 Experimental methods and reproducibility

All tests use Xschem-exported netlists and the SKY130 `sky130_fd_pr_{n,p}fet_01v8` MOS models in ngspice. Process corners are TT, SS, FF, SF, and FS, with supplies 1.62/1.80/1.90 V and temperatures $-40/27/125\,^{\circ}$C. The standalone comparator uses both $\Delta V_{\mathrm{in}} = \pm 10$ mV polarities, giving $5 \times 3 \times 3 \times 2 = 90$ PVT tests. NAND2, standalone SR latch, and the integrated hierarchy each cover $5 \times 3 \times 3 = 45$ combinations with internal multi-state excitation.

Stage 9D drives four consecutive 4-MHz cycles with $\Delta V_{\mathrm{in}} = +10, -10, +10, -10$ mV. Comparator outputs $V_{out,P/N}$ and stored outputs $Q/\overline{Q}$ are checked at eleven time points per case, including input polarity reversal during comparator reset. Each of the four observed outputs uses a 20-fF load. Stage 9E examines five critical PVT configurations in more detail: HOLD is verified by window MIN/MAX, and supply energy is integrated over cycles 2–4, excluding startup.

The software-defined logic screening criteria are HIGH $\geq 0.8V_{\mathrm{DD}}$ and LOW $\leq 0.2V_{\mathrm{DD}}$. These values are verification thresholds rather than a full final-system noise-margin guarantee. Stage 9D and 9E netlist SHA-256 values are checked for identity. All numerical result tables trace back to CSV files included alongside the LaTeX source.

6 Standalone comparator results: testing regeneration theory

6.1 PVT behavior and timing bound

Stage 8 produced **90/90 functional PVT passes**. At TT, 1.8 V, 27 °C and $\Delta V_{\mathrm{in}} = \pm 10$ mV, the measured t_{dec} was 1.790 614 ns, $P_{\mathrm{avg}} = 1.207\,488$ µW, and energy per clock period was 0.301 872 pJ. Over 90 cases, maximum delay was 2.523 052 ns at FS/1.62 V/125 °C, while maximum average power was 2.822 138 µW at FF/1.90 V/125 °C. All results are pre-layout and use ideal differential voltage sources.

6.2 Quantitative test of the logarithmic trend

At TT/1.8 V/27 °C, the four absolute differential-input values lead to the measurements below. Motivated by Eq. (5), we fit the empirical relation

$$t_{\mathrm{dec}} = a + b \ln\left(\frac{10\,\mathrm{mV}}{|\Delta V_{\mathrm{in}}|}\right). \tag{11}$$

Least-squares fitting yields

$$a = 1.797\,34\,\text{ns}, \qquad b = 0.100\,65\,\text{ns}, \qquad R^2 = 0.99257. \tag{12}$$

This supports the predicted increase in decision time as differential input shrinks. The slope b is *empirical* for this stimulus and is not a direct extraction of C_{eq}/g_m across the entire PVT space. Input levels of 1–2 mV are deterministic characterizations, not offset-safe guarantees.

6.3 Clock-slew stress study

Before adopting 5-ns clock edges, 11-transistor terminal screening examined 1-, 2-, and 5-ns transitions (5-ps maximum timestep in refined runs). The peak internal P/Q voltage at TT/1.90 V/$-40\,^\circ$C dropped from 2.04257 V to 1.99456 V and then 1.94099 V. The experiment demonstrates a clock-edge impact on simulated overshoot. However, residual $|V_{GB}|$ stress around 1.963–1.965 V on M3/M4 persisted at 5 ns.

$\lvert\Delta V_{in}\rvert$ (mV)	Delay (ns)	Power (μW)	Status
10	1.790614	1.207488	PASS
5	1.872678	1.233943	PASS
2	1.968739	1.260530	CHAR.
1	2.020851	1.275510	CHAR.

Table 4: StrongARM sensitivity at TT/1.80 V/27 C.

Edge (ns)	P/Q peak (V)	$\lvert V_{GB}\rvert$ M3/M4 (V)
1	2.042568	1.991906
2	1.994561	1.977968
5	1.940994	1.963173

Table 5: Clock-slew screening at TT/1.90 V/-40 C.

7 From NAND2 to SR latch: standalone verification

7.1 NAND2: correct Boolean behavior, asymmetric delay

A NAND2 gate with a 20-fF load passed **45/45 PVT cases**. At TT/1.8 V/27 °C, falling delay was 67.0083 ps, while rising delay was 99.6120 ps. This direction is consistent with differing effective pull-up/pull-down strengths, although a first-order ON-resistance formula does not independently predict the precise times.

The nominal average power during a 40-ns stimulus was 1.930 932 µW, yielding the *test-window energy*

$$E_{40} = P\Delta t = (1.930932 \times 10^{-6})(40 \times 10^{-9})\,\text{J} = 77.2373\,\text{fJ}. \tag{13}$$

This is not energy per decision for the 4-MHz system.

7.2 SR latch: SET, RESET, and HOLD from the cross-coupled network

The eight-transistor SR latch passed all **45/45 PVT cases**. Each simulation examined eight SET/HOLD/RESET/HOLD phases repeated twice, yielding 16 Q/$\bar{\text{Q}}$ level values and four delay measurements per case. At nominal conditions, $t_{\text{SET}\to Q} = 113.585$ ps and $t_{\text{SET}\to\overline{Q}} = 209.395$ ps; worst-case delay was 310.692 ps at SS/1.62 V/$-40\,^\circ$C.

Nominal latch power over its 75-ns stimulus was 4.047 804 µW, giving $E_{75} = 303.5853$ fJ for the entire window. It cannot be compared directly as *energy per decision* against the 9D/9E measurements because pulse sequences and integration windows differ. The actual standalone nominal simulator log used before PVT automation is reproduced in Fig. 1.

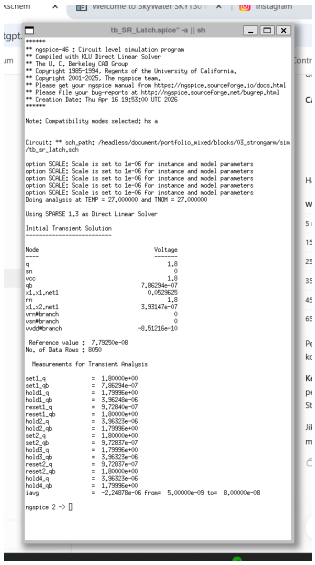


Figure 1: User-supplied ngspice log screenshot for the standalone SR latch. Its measurement values were subsequently checked over the PVT dataset.

Circuit	PVT	Delay 1 (ps)	Delay 2 (ps)	Power (μW)
NAND2	45/45	67.0083	99.6120	1.930932
SR Latch	45/45	113.585	209.395	4.047804

Table 6: Nominal standalone NAND and latch; their delay definitions differ.

8 Integration: four successive decisions and reset retention

The integration connects $V_{out,P}$ directly to S_N and $V_{out,N}$ to R_N. The Xschem circuit contains hierarchical StrongARM 11T and SR-latch 8T instances, implementing *19 transistor instances in total*. The original user-submitted schematic is shown on the cover and in Fig. 2.

Stage 9D passed **45/45 PVT configurations**, checking 11 time samples per configuration (495 logic checks). At TT/1.8 V/27 °C, integrated decision-path delay was 2.438 500 ns and average power was 1.761 937 µW. Worst delay was 3.189 130 ns at SS/1.62 V/125 °C; maximum average power was 4.922 463 µW at FF/1.90 V/125 °C.

Integrated delay must not be decomposed by directly adding standalone comparator and latch delay results, since loads, stimulus transitions, and measurement definitions differ. The numerical nominal difference $2.438500 - 1.790614 = 0.647\,886$ ns compares *two different test setups*; it is not a direct extraction of intrinsic latch delay.

HOLD after SET demonstrates that $Q \approx V_{\mathrm{DD}}$ even though both comparator outputs reset HIGH. Reversing the input polarity during reset does not change the stored Q until the next evaluation. The four input signs $+10, -10, +10, -10$ mV correctly create HIGH, LOW, HIGH, LOW at Q.

9 Energy, power, and 4-MHz timing checks

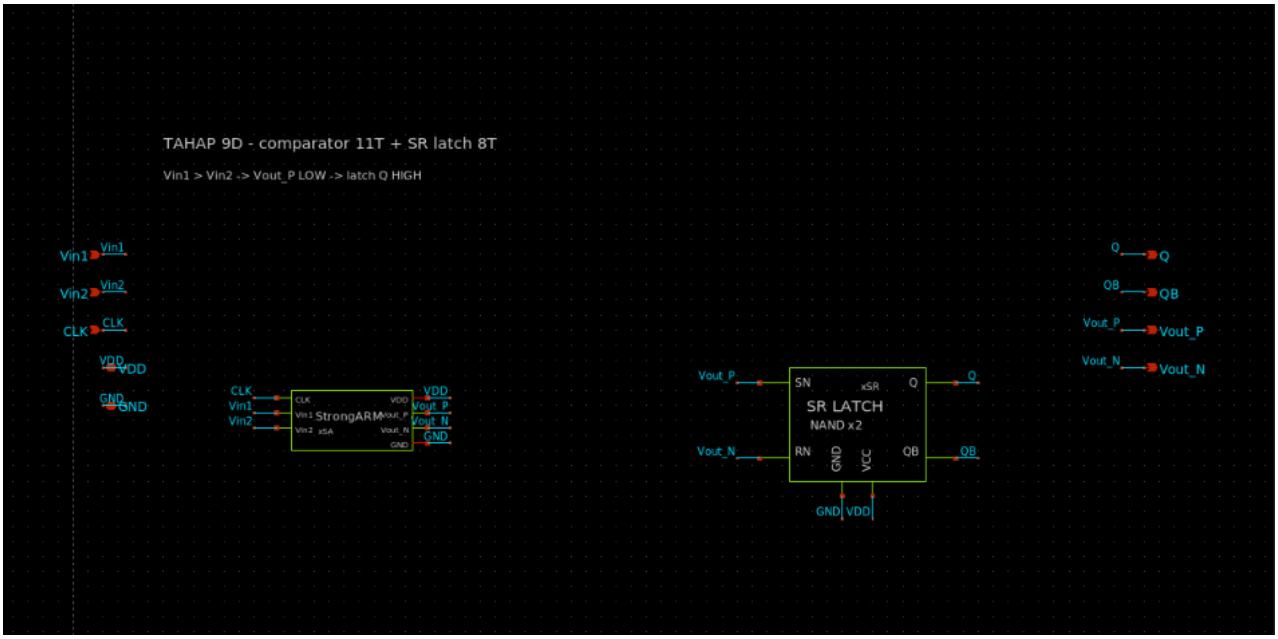


Figure 2: The submitted Xschem view of StrongARM plus the cross-coupled NAND SR latch; net labels implement the interblock connections.

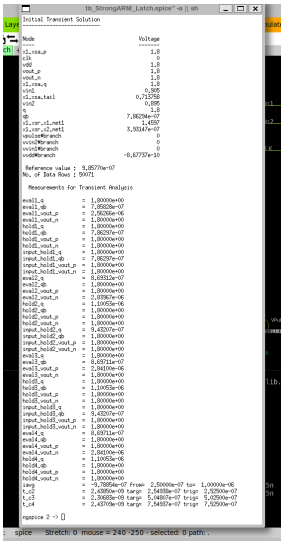


Figure 3: Actual ngspice measurement log for the integrated nominal test. This is a screenshot of reported `.meas` values, not a fabricated waveform.

Corner	Max delay (ns)	Max power (μW)	PASS
TT	2.85466	2.88722	9/9
SS	3.18913	2.36672	9/9
FF	2.63117	4.92246	9/9
SF	2.58865	2.58022	9/9
FS	3.14921	3.51772	9/9

Table 7: Integrated StrongARM+SR-latch PVT summary.

9.1 Cycle-energy identity

Supply energy over a single clock cycle is defined by

$$E_k = -V_{\mathrm{DD}} \int_{(k-1)T_s}^{kT_s} I_{\mathrm{VDD}}(t)\, dt. \tag{14}$$

For constant supply voltage and a steady-state sequence,

$$\overline{E}_{\mathrm{cycle}} = \frac{\overline{P}}{f_s} = \overline{P}\, T_s. \tag{15}$$

Substituting Stage 9D measurements,

$$E_{\mathrm{nom,9D}} = (1.761937\,\mu\mathrm{W})(250\,\mathrm{ns}) = 0.440\,484\,\mathrm{pJ}, \tag{16}$$

$$E_{\mathrm{worst,9D}} = (4.922463\,\mu\mathrm{W})(250\,\mathrm{ns}) = 1.230\,616\,\mathrm{pJ}. \tag{17}$$

Stage 9E independently integrates the supply current for cycles 2–4 and reports a worst average of 1.230 635 pJ per steady-state cycle. Its difference from 1.230616 pJ is approximately 0.0015%, which is consistent with different integration windows. This checks the *energy identity* numerically, not the completeness of transistor-level device physics.

For a single 20-fF output capacitor, the ideal supply energy for one LOW-to-HIGH charge is

$$E_{C,\mathrm{supply}} = C_L V_{\mathrm{DD}}^2 = (20 \times 10^{-15})(1.8)^2 = 64.8\,\mathrm{fJ}. \tag{18}$$

One-half, 32.4 fJ, remains stored in the capacitor and the other half is ideally dissipated during charging. This is *not* the system energy estimate: there are four loaded outputs, additional parasitics, regenerative supply current, short-circuit conduction, and stimulus-dependent transition activity.

9.2 Timing margin: necessary but not sufficient

With a 250-ns period and worst measured delay of 3.18913 ns, the delay/period ratio is 1.276%. Relative to an idealized evaluation pulse of approximately 125 ns, it occupies 2.55%. These ratios suggest substantial timing margin within the functional testbench. They do not certify timing closure for the complete DAC–integrator–quantizer chain, where nonoverlapping clocks, setup/hold, and extracted parasitics still matter.

9.3 Window-wide HOLD audit

Stage 9E measures the minimum and maximum values of Q and $\overline{Q}$ throughout all HOLD windows in five critical configurations, including a differential-input transition while the comparator remains in reset. All **5/5** pass. The maximum individual 250-ns-cycle energy measured over cycles 2–4 was 1.230 673 pJ.

Corner	V_{DD}	T (°C)	Delay (ns)	E (pJ)	Peak (V)	Result
TT	1.80	27	2.43850	0.440493	1.85358	PASS
SS	1.62	125	3.18913	0.381292	1.66466	PASS
FF	1.90	125	2.52042	1.230635	1.94679	PASS
SF	1.90	27	2.05609	0.496674	1.96053	PASS
TT	1.90	-40	2.23958	0.456237	1.95904	PASS

Table 8: Five critical 9E results.

10 Voltage limits: functional success is not reliability signoff

The separate terminal-stress audit computed six differential terminal voltages (GS, GD, DS, GB, DB, SB) for each MOSFET. A 1.95-V absolute-magnitude criterion was used for screening, but formal model limits depend on the *terminal pair and polarity*. Official SKY130 1.8-V NMOS documentation specifies $0 \le V_{DS} \le 1.95\,\text{V}$, $0 \le V_{GS} \le 1.95\,\text{V}$, and $-1.95\,\text{V} \le V_{BS} \le 0.3\,\text{V}$; PMOS V_{DS}/V_{GS} limits have the opposite sign and V_{BS} extends from $-0.1\,\text{V}$ to $+1.95\,\text{V}$ [2].

A 5-ns clock significantly reduced internal-node overshoot compared with 1-ns edges. Nevertheless, M3/M4 exhibited a maximum $|V_{GB}| \approx 1.964\,69\,\text{V}$ at SF/1.90 V/27 °C in the refined screen. Stage 9E also found an output-*node* maximum of 1.960 53 V and minimum $-0.002\,166\,\text{V}$. A node peak is not identical to a gate-oxide or terminal-pair stress value: these data do not establish physical device failure. Equally, functional PASS does not establish device-model or long-term reliability signoff.

Available results do not include post-layout parasitic extraction, long-term voltage stress, validated TDDB/HCI degradation models, production yield, or a final full-terminal audit of the integrated design. If 1.90 V must belong to the guaranteed operating range, the flagged conditions require legitimate PDK/foundry justification or a design remedy. Consequently, the device-reliability status remains *not signed off*.

11 Does theory agree with the simulation?

The scientific question is not whether all numerical values must coincide; it is whether a theoretical model makes testable predictions and whether the measured results corroborate them under stated conditions. The evidence has three levels.

First: exact agreement with Boolean relations. NAND and cross-coupled latch equations predict SET/RESET/HOLD and $V_{in1} > V_{in2} \Rightarrow Q = 1$. All 45 Stage-9D configurations and the five Stage-9E critical tests agree under the stated digital thresholds. This matches discrete predictions to thresholded analog transistor simulation.

Second: trend agreement, not an absolute delay prediction. The regeneration model says t_{dec} rises as $|\Delta V_{\text{in}}|$ falls. The measured 10-mV delay of 1.7906 ns increases to 2.0209 ns at 1 mV, and the empirical $a + b\ln(10\,\text{mV}/|\Delta V_{\text{in}}|)$ model gives $R^2 \approx 0.993$. Without extracted $g_{m,\text{reg}}$ and C_{eq}, one cannot independently derive the exact 1.7906-ns figure solely from W/L.

Third: mathematical agreement in the energy identity. $E = P/f_s$ maps maximum 9D power to 1.230616 pJ; cycle-by-cycle numerical integration in 9E gives 1.230635 pJ. The difference is approximately 0.0015%, but these are *not independent transistor-physics models*: both rely on related SPICE current waveforms.

Ideal Boolean CMOS theory does not require transient node voltages to remain strictly between the supply rails. The 1.96053-V peak under 1.90-V supply illustrates capacitive coupling and transient effects already visible before physical layout. It is a reliability question to be investigated, not a contradiction of the NAND truth table.

12 Block closure and unresolved engineering work

The 19-MOSFET hierarchy correctly captures, updates, and retains a one-bit decision across resets for the tested four-cycle pattern. Transistor-level functionality, 45 integrated PVT configurations, and the continuous-HOLD/steady-cycle-energy audit were completed. Block 3 may therefore be marked **complete for pre-layout schematic functional verification**.

The remaining path toward silicon readiness includes justification of the 1.90-V device operat-

Aspect	Theory	Evidence	Finding
NAND/SR logic	SET/RESET/HOLD	45/45 + 5/5	MATCH
Regeneration	Delay rises as input falls	$R^2 = 0.99257$	TREND
Energy	$E = P/f_s$	1.230616 vs 1.230635 pJ	MATCH
Absolute delay	Needs g_m, C	Not extracted	OPEN
Reliability	Terminal-specific limits	1.96469 V flag	OPEN

Table 9: Explicit assessment of theory versus evidence.

ing envelope, post-integration mismatch Monte Carlo, realistic source impedance and kickback in the modulator loop, stochastic noise/metastability characterization, layout DRC/LVS, and parasitic extraction with post-layout verification. The next intended architecture stage is Block 4, a one-bit feedback DAC; independent block success does not automatically establish full delta-sigma loop performance.

12.1 Data provenance and reproduction limits

The archive contains the source CSV files underlying Stages 8/9A/9C/9D/9E and the clock-slew study. Figures on the cover and within the report use user-supplied screenshots only; no waveform has been synthesized for presentation. The detailed prior Monte Carlo and kickback raw campaigns were unavailable in the report inputs, so they are not included in final signoff tables. Any theory requiring internal g_m, capacitances, or Pelgrom coefficients remains symbolic and distinct from SPICE measurement results.

A Detailed PVT evidence

The following table lists all 45 Stage-9D integrated PVT combinations. The full source CSV including eleven logic samples per case is included in the `data/` directory.

Table 10: All 45 integrated PVT observations.

Corner	V_{DD}	T	Delay (ns)	Power (μW)	E (pJ)	Status
TT	1.62	-40	2.7227	1.2572	0.3143	PASS
TT	1.62	27	2.7429	1.3440	0.3360	PASS
TT	1.62	125	2.8547	1.7656	0.4414	PASS
TT	1.80	-40	2.3842	1.5884	0.3971	PASS
TT	1.80	27	2.4385	1.7619	0.4405	PASS
TT	1.80	125	2.6653	2.4270	0.6067	PASS
TT	1.90	-40	2.2396	1.8249	0.4562	PASS
TT	1.90	27	2.3337	2.0723	0.5181	PASS
TT	1.90	125	2.6277	2.8872	0.7218	PASS
SS	1.62	-40	3.1071	1.2638	0.3159	PASS
SS	1.62	27	3.1208	1.3287	0.3322	PASS
SS	1.62	125	3.1891	1.5251	0.3813	PASS
SS	1.80	-40	2.7478	1.5715	0.3929	PASS
SS	1.80	27	2.7742	1.6839	0.4210	PASS
SS	1.80	125	2.9134	2.0188	0.5047	PASS
SS	1.90	-40	2.5904	1.7734	0.4434	PASS
SS	1.90	27	2.6339	1.9344	0.4836	PASS
SS	1.90	125	2.8363	2.3667	0.5917	PASS

Corner	V_{DD}	T	Delay (ns)	Power (μW)	E (pJ)	Status
FF	1.62	-40	2.3849	1.2660	0.3165	PASS
FF	1.62	27	2.4214	1.4231	0.3558	PASS
FF	1.62	125	2.6312	2.7614	0.6903	PASS
FF	1.80	-40	2.0739	1.6661	0.4165	PASS
FF	1.80	27	2.1932	1.9742	0.4935	PASS
FF	1.80	125	2.5283	4.0232	1.0058	PASS
FF	1.90	-40	1.9626	1.9763	0.4941	PASS
FF	1.90	27	2.1389	2.3825	0.5956	PASS
FF	1.90	125	2.5204	4.9225	1.2306	PASS
SF	1.62	-40	2.4542	1.2380	0.3095	PASS
SF	1.62	27	2.4859	1.3107	0.3277	PASS
SF	1.62	125	2.5887	1.5854	0.3964	PASS
SF	1.80	-40	2.1080	1.5522	0.3880	PASS
SF	1.80	27	2.1699	1.6982	0.4245	PASS
SF	1.80	125	2.3837	2.1694	0.5424	PASS
SF	1.90	-40	1.9590	1.7701	0.4425	PASS
SF	1.90	27	2.0561	1.9867	0.4967	PASS
SF	1.90	125	2.3363	2.5802	0.6451	PASS
FS	1.62	-40	3.0492	1.2741	0.3185	PASS
FS	1.62	27	3.0390	1.3817	0.3454	PASS
FS	1.62	125	3.1492	2.1373	0.5343	PASS
FS	1.80	-40	2.7124	1.6259	0.4065	PASS
FS	1.80	27	2.7445	1.8333	0.4583	PASS
FS	1.80	125	2.9770	2.9531	0.7383	PASS
FS	1.90	-40	2.5727	1.8828	0.4707	PASS
FS	1.90	27	2.6548	2.1674	0.5419	PASS
FS	1.90	125	2.9487	3.5177	0.8794	PASS

B Data provenance and verification boundaries

The Stage-8 comparator dataset contains 90 PVT entries (two polarities) plus eight small-input characterization points. Standalone NAND and SR-latch records each contain 45 PVT entries. Stage 9D has 45 integration cases and Stage 9E has five critical ones. Stage-9D/9E netlist SHA-256 values match.

```
SHA256 (9D/9E):
59c7e5926287c5d9c447361c007ae862f0cfe3ee18089a0f0f7d7fa241482cbc
```

Source caveat: the Stage-8 summary warns that an imported TT log, if used, cannot independently prove its clock transition time. The reported 90/90 PASS follows the supplied validation dataset and remains subject to that provenance qualification. No claim is made that the authoring environment independently reran all transistor simulations.

References

[1] B. Razavi, “The StrongARM Latch [A Circuit for All Seasons],” *IEEE Solid-State Circuits Magazine*, vol. 7, no. 2, pp. 12–17, 2015. DOI: 10.1109/MSSC.2015.2418155.

[2] SkyWater Technology, *SKY130 PDK: Device Details, 1.8-V MOSFET Model Validity*. https://skywater-pdk.readthedocs.io/en/main/rules/device-details.html.

[3] J. M. Rabaey, A. Chandrakasan, and B. Nikolić, *Digital Integrated Circuits: A Design Perspective*, 2nd ed., Prentice Hall, 2003.

[4] M. J. M. Pelgrom, A. C. J. Duinmaijer, and A. P. G. Welbers, "Matching properties of MOS transistors," *IEEE Journal of Solid-State Circuits*, vol. 24, no. 5, pp. 1433–1439, 1989.

[5] User-supplied Xschem/ngspice simulator outputs and CSV data (Stages 8, 9A, 9C, 9D, 9E), October 2026, distributed with this report in `data/`.